

Interactive Design with Autistic Children Using LLM and IoT for Personalized Training: The Good, The Bad and The Challenging

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ABSTRACT

The advent of generative artificial intelligence technologies, such as Large Language Models (LLMs) and Large Vision Models (LVMs), has shown promising results in both academic and industrial sectors, leading to widespread adoption. However, there has been limited focus on applying these technologies to assist children with special needs like Autism Spectrum Disorder (ASD). Meanwhile, conventional personalized training with interactive design for children with special needs continues to face significant challenges with traditional approaches. This workshop aims to provide a platform for researchers, software developers, medical practitioners, and designers to discuss and evaluate the benefits and drawbacks of using LLMs and the Internet of Things (IoT) for the diagnosis and personalized training of autistic children. Through a series of activities, including oral presentations, demonstrations, and panel discussions, this half-day workshop seeks to foster a network of experts dedicated to improving the lives of children with special needs and to inspire further research on leveraging emerging ubiquitous technologies for these underprivileged users, their caregivers, and special education teachers.

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CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools; Ubiquitous and mobile computing design and evaluation methods.

KEYWORDS

Autism, Children, Ubiquitous Computing, Large Language model (LLM), Interaction Design, Personalized Training

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1 BACKGROUND

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder with a median prevalence of 100/10,000 worldwide as reported in [16]. It specifically influences how individuals interact with others, communicate, learn, and behave. The symptoms of ASD (e.g., lack of eye contact, repetitive behaviors, etc.) generally appear in the first two years of life. Though recent advantages in ASD assistive and interactive technologies have been made to help the early intervention and treatment of autistic children, including some of our works [10–13], difficulties in trust building among stakeholders, effective communication with autistic children, building a natural and playful experiment environment, translating autistic children's "irrational" input, and collecting oral feedback from children persist in interactive design with autistic children [4].

The Internet of Things (IoT), enabled by ubiquitous computing power, brings us new opportunities in ubiquitous health and human monitoring. Smart devices equipped with numerous sensors, including contactless wireless sensors, facilitate the discreet tracking of human activities [9] and can passively gather behavioral and physiological data in everyday settings. These devices enable applications like fall detection [6], vital sign monitoring [15] and activity recognition [7], ensuring maximum comfort and convenience for users during data collection. This way, IoT-based sensing networks integrated in natural settings (such as home and regular special education classrooms) could offer the capability of actively collecting rich behavioral and physiological data from the children in a ubiquitous manner without distributing the autistic children conducting the daily training or activities.

Additionally, the emergence of generative foundation models such as the Large Language Model (LLM) and Large Vision Model (LVM) has presented promising outcomes in both academic and industrial domains [1, 17]. However, there is little work focused on combining LLM with underrepresented groups like autism, to list a few: [2, 3, 8], which motivates the proposed workshop. Combining with IoT technology, the extracted children's behaviors/habits from the IoT sensing network could further feed to a customized LLM to generate corresponding personalized training content for individual children, which aims to overcome the aforementioned challenges raised in mainstream typical interactive design.

The goal of this workshop is to provide a platform for researchers, software and medical practitioners, and designers to share and debate both the pros and cons of applying the Large Language Model (LLM) and Internet of Things (IoT) for diagnosis and personalized training for autistic children. Through hosting multiple activities during the half-day workshop, including oral presentation, demo, and panel discussion, we hope to use this opportunity to build a network of experts to dedicate our efforts to benefiting children with special needs and further inspire the research on taking advantage of the emerging technologies for these under-privileged group users, caregivers, and special education teachers.

Additionally, we are particularly interested in calling for experts and practitioners in the Asia-Pacific region to attend this event, as more and more focus has been on Chinese children with autism in the past ten years[5, 10–14].

2 ORGANIZERS

Yongfu Wang is a PhD student in the University of Nottingham Ningbo China Computer Science program. He received a master's degree in Computer Science from Northwestern University (US) and a bachelor's degree in Computer Science with a minor in Mathematical Science at Wenzhou-Kean University. His PhD research interest focuses on utilizing ubiquitous wearable sensors and Internet-of-Things (IoT) devices with haptic sensing for healthcare. His prior work was published at ACM CHI, CSCW, and UbiComp conferences.

Mingyue Tang is a Ph.D. student in the Department of Computer Science at the University of Illinois Urbana-Champaign (UIUC). She received her master's degree from the University of Virginia (UVA) Link Lab. Her research interests include wireless sensing, mobile computing, signal processing, and the Internet of Things

(IoT) in health. Through her work, she aims to create innovative solutions that can address real-world problems and make a positive impact on society. Her prior work was published at ICLR, KDD, IPSN, SenSys, etc. conferences, and ACM Transactions on Sensor Networks (TOSN) and SPJ Health Data Science journals.

Yifan He is an assistant professor at the Department of Software Engineering, Zhejiang University of Finance and Economics. He received his Ph.D. and Master's degree at the University of Tsukuba, Japan in 2023 and 2020, respectively. Currently, his research interests include evolutionary optimization and learning, and evolvable hardware. He aims to build an evolvable and intelligent problem-solving system. Previously, he received his Bachelor's degree at Wenzhou-Kean University, China in 2017, where he conducted several studies in HCI and assistive technology. His prior work was published at GECCO, HCII conferences and Operations Research Perspectives journals, etc. He is a reviewer of several journals and conferences, such as Information Sciences, IEEE Trans. on Evolutionary Computation, and ALIFE, etc.

Tiffany Y. Tang is an associate professor at Wenzhou-Kean University. In May 2015, she co-founded the Center on Innovation and Assistive Technology for Autism & Neurodevelopmental Disorders at Wenzhou-Kean University (then called Wenzhou-Kean Autism Research Center). Till now, her team has published a total of 26 book chapters, journal and conference papers related to innovative technology for autism. Dr. Tang's most recent research interests include affective computing and computational sensing on autism, sensing and wearable technologies for human-computer/robot interaction, human-factored recommendation systems, assistive and wearable technology for special education, data mining and artificial intelligence in education.

All co-organizers have been affiliated with the Center on Innovation and Assistive Technology for Autism & Neurodevelopmental Disorders at Wenzhou-Kean University since 2015.

3 PRE-WORKSHOP PLANS

We are expecting the main target participants of the proposed workshop to include researchers, designers, practitioners and children who are currently working on or are interested in the study involving LLM, LVM, IoT, and/or Smart Sensors for personalized training design for autistic children in an interactive approach. The various channels will be used to attract potential participants, including the mailing lists (e.g., CHI-kids mailing list), social media (e.g., Tweet, Facebook, and YouTube), university websites, and conferences (e.g., DIS and CHI). The workshop website (<https://idwac.github.io/>) will also be used for promotion purposes.

4 WORKSHOP STRUCTURE

This half-day (3-4 hours long) open workshop will serve as a collective platform that brings together researchers, practitioners, designers, and children with diverse cultures and backgrounds to share their recent works, applications, and creative ideas that contribute to the area focused on interactive design with autistic children using LLM, LVM, and/or IoT for personalized training. The proposed workshop structure is as follows:

- (1) **Welcome and Introduction (15 minutes):** The facilitator (one of the co-organizers) will introduce the theme of

the workshop: *Interactive design with autistic children using LLM and IoT for personalized training: the good, the bad and the challenging* followed by sharing the workshop agenda, addressing challenges and objectives.

- (2) **Keynote Speech (25 minutes):** The invited keynote speaker will deliver a talk on the topic of emerging Technology-Based Intervention (TBI) research trends for children with autism and other neurodevelopmental Disorders.
- (3) **Short Presentations (80 minutes):** We are expecting to have up to 8 groups of participants, and each group will give a 7-minute presentation on their position papers and work-in-progress reports, followed by a 3-minute short Q&A session.
- (4) **Demos and Coffee Break (60 minutes):** Up to 10 groups of participants present their demos or applications in an open setting related to our workshop theme, including leading-edge systems in use, under development or at a testing stage and art installations such as prototypes, devices, games, and videos.
- (5) **Open Discussion and Brainstorm (30 minutes):** Discussion on the current research works/outcomes, trends and challenges of involving LLM and IoT in interactive design with autistic children for assisting in personalized learning for autistic children and brainstorming the potential future directions of the future works.
- (6) **Closing and wrap-up (20 minutes):** Recap of workshop key points; Gratitude and acknowledgments.

5 TYPES OF SUBMISSION AND SELECTION CRITERIA

This workshop explores the benefits, challenges, and future directions for involving creative interactive design using LLMs and IoT with/for autistic children in diagnosis and personalized training. By engaging in presentations, demonstrations, and group discussions, participants will have the chance to exchange their related experiences and insights.

Submissions of *position papers*, *work-in-progress reports*, or *demonstration papers* for a short presentation or demonstration related to the interactive design with autistic children using LLM and IoT for diagnosis and personalized training or relevant fields are welcomed. Specifically, the proposed workshop is expecting:

- (1) **Position papers (2-4 pages)** discussing research questions, opportunities, benefits, or challenges
- (2) **Work-in-progress reports (2-4 pages)** highlighting current research
- (3) **Demonstration papers (1 page)** illustrating a leading-edge system in use, under development, or at a testing stage

The suggested topics include (but are not limited to):

- (1) Optimized and Personalized Training for Special Education
- (2) AI, IoT, and/or Smart Sensors for Special Education
- (3) Large Language Models (LLMs) and/or Large Vision Models (LVM) for Special Education
- (4) Technology-Based Intervention (TBI) for Special Education
- (5) Interactive Design with Children
- (6) Emerging Applications for Special Education or Healthcare

All submitted papers will be reviewed and evaluated by a panel of a minimum of two reviewers, mainly based on originality, technical soundness, relevance, and potential contribution to the autism/special education community. The workshop organizers will be in charge of overseeing the coordination of the review process. Accepted works will be invited to present their submissions in a dedicated presentation or demo session.

6 EXPECTED ATTENDANCE

The proposed workshop is planned to run in an open format to provide a platform for researchers, software and medical practitioners, and designers to share and debate both the pros and cons of applying the Large Language Model (LLM) and the Internet of Things (IoT) for diagnosis and personalized training for autistic children. We hope to use this opportunity to build a network of experts to dedicate our efforts to benefiting children with special needs and further inspire the research on taking advantage of the emerging ubiquitous technologies for these underprivileged group users, caregivers, and special education teachers.

Anticipated attendees include authors of accepted papers, UbiComp participants, invited speakers, and workshop assistants.

7 POST-WORKSHOP PLANS

To continue the long-term impact of the workshop and the creation of tangible outcomes, the following follow-up activities will be arranged: synchronizing all works (position papers, work-in-progress reports, and demonstration papers) presented in the proposed workshop and discussing outcomes on our workshop website after obtaining written permission; inviting workshop participants to publish their works in a selected journal special issue.

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Appendices

A CALL FOR PARTICIPATION

UbiComp 2024 Workshop: *Interactive Design with Autistic Children using LLM and IoT for Personalized Training: The Good, The Bad and The Challenging*

Important Dates:

- (1) Submission Deadline: June 30, 2024 AoE
- (2) Acceptance Notification: July 10, 2024 AoE
- (3) Camera-ready deadline: July 26, 2024 AoE

- (4) Submission website: (Via UbiComp PCS. Track: UbiComp/ISWC 2024 Workshop on Design with Autistic Children.)

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- (4) Technology-Based Intervention (TBI) for Special Education
- (5) Interactive Design with Children
- (6) Emerging Applications for Special Education or Healthcare

Accepted works will be invited to present their submissions in a dedicated presentation or demo session.